

INFO 5613

Network Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (67.0 / 100)	54.5 / 100

54

CURRENT AVG SCORE
across 6 job roles

92

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces theories and methods for analyzing relational data in social, information, and other complex networks. Students will understand the processes and theories explaining network structure and dynamics as well as develop skills analyzing and visualizing real-world network data. No math or statistics training required, but course will assume familiarity with Python.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	67.0	100.0	+33.0	HIGH
ML Engineer	58.5	85.6	+27.1	MEDIUM
Machine Learning Engineer	56.2	88.3	+32.2	HIGH
AI Engineer	53.6	98.4	+44.9	HIGH
Applied Scientist	52.0	100.0	+48.0	HIGH
MLOps Engineer	39.5	77.7	+38.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Data Scientist 67.0 → 100.0 (+33.0 pts)

sets analytics build people problems approaches

ML Engineer 58.5 → 85.6 (+27.1 pts)

media learning machine systems models model world build

Machine Learning Engineer 56.2 → 88.3 (+32.2 pts)

sets learning analytics design science models model applications

AI Engineer 53.6 → 98.4 (+44.9 pts)

models learning build software technical model systems teams

Applied Scientist 52.0 → 100.0 (+48.0 pts)

biology computational research reasoning interdisciplinary problems models technical

MLOps Engineer 39.5 → 77.7 (+38.2 pts)

model text systems learning machine media models error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.